Email Forensic Analysis Report

Case EFT-20261028-78D60AAB - generated 2026-10-28T06:55:57+00:00

Case identifier	EFT-20261028-78D60AAB
Analysis time (UTC)	2026-10-28T06:55:57.790720+00:00
Report generated (UTC)	2026-10-28T06:55:57+00:00
Tool	Email Forensics Toolkit v1.0.0
Source	S06_idn_homograph.eml
Input method	file upload
Evidence size	867 bytes
SHA-256 (at intake)	d493c308f60d28f3bfe50dc28ac8bae53f19fc70b2a5277f39da3abbf68b408d
Integrity re-check	INTACT - stored original matches the intake hash (verified 2026-10-28T06:55:57+00:00)
Analysis settings	scoring=severity_sum; trusted domains=['uni-example.edu']; trusted authserv-ids=auto; extra protected domains=none; DNS lookups=off; clock-skew tolerance=5 min

1. Risk assessment

Category: **HIGH** Score: **9** Approach: severity_sum (thresholds medium ≥ 3, high ≥ 6)

The score ranks how many and how serious the header indicators are. It is not a calibrated probability of maliciousness, and a low score does not prove a message is safe (e.g. a compromised legitimate account passes every header check).

Rule	Finding	Severity	Points	Why these points
DOMAIN-LOOKALIKE	From domain resembles paypal.com (idn-homograph)	high	6	high severity = 6 point(s)
SENDER-DISPLAYNAME-BRAND	Display name invokes 'paypal' but the address is not on paypal.com	medium	3	medium severity = 3 point(s)

2. Extracted header information

From	"PayPal" <security@xn--pypal-4ve.com>
Sender	-
Reply-To	-
Return-Path	<security@xn--pypal-4ve.com>
To	<student@uni-example.edu>
Cc	-
Subject	Verify your identity
Date	Mon, 14 Sep 2026 09:29:00 +0000 (parsed: 2026-09-14T09:29:00+00:00)
Message-ID	<707213889216.816252@xn--pypal-4ve.com>
Domains observed	xn--pypal-4ve.com
IP addresses	10.20.0.11 (private), 45.137.22.15 (public)
Header fields	11 total; 2 Received; full message

3. Apparent delivery route (earliest first)

Trusted infrastructure: uni-example.edu. Origin IP seen by trusted server: 45.137.22.15 (public). Hops marked 'claimed' were written by servers outside the trust boundary and may be forged.

#	From (HELO / rDNS / IP)	By	Timestamp	Delay	Trust
1	mx.xn--pypal-4ve.com / mx.xn--pypal-4ve.com / 45.137.22.15	mx1.uni-example.edu	2026-09-14T09:30:00+00:00	-	trusted
2	mx1.uni-example.edu / mx1.uni-example.edu / 10.20.0.11	mailbox01.uni-example.edu	2026-09-14T09:30:12+00:00	+12 s	trusted

4. Authentication results

Mechanism	Status	Reported	Domain	Interpretation
SPF	pass	pass	xn--pypal-4ve.com	The connecting IP is authorised by the SPF record of the envelope sender (Return-Path) domain. SPF says nothing about the visible From.

Mechanism	Status	Reported	Domain	Interpretation
DKIM	**pass**	pass	xn--pypal-4ve.com	A DKIM signature verified, so the signed headers/body were not altered after signing by the signing domain (d=).
DMARC	**pass**	pass	xn--pypal-4ve.com	SPF or DKIM passed for a domain aligned with the visible From domain.

Alignment (computed): From=xn--pypal-4ve.com, SPF domain=xn--pypal-4ve.com (aligned: True), DKIM pass domains=xn--pypal-4ve.com (aligned: True). Relaxed alignment computed by this tool from the trusted results; it approximates, but is not, a DMARC evaluation.

Trusted authserv-ids: mx1.uni-example.edu; untrusted results ignored: 0.

5. Findings

1. From domain resembles paypal.com (idn-homograph) [HIGH] DOMAIN-LOOKALIKE

Evidence: From: xn--pypal-4ve.com

Evidence: 'p■ypal.com' uses non-Latin characters (CYRILLIC, LATIN) that render like 'paypal'.

Why it matters: Look-alike domains are registered so that a hurried reader mistakes them for a trusted organisation. Authentication can fully pass for them because the attacker owns the domain.

Limitations: Similarity is heuristic; a protected organisation may own the look-alike (defensive registration).

2. Display name invokes 'paypal' but the address is not on paypal.com [MEDIUM] SENDER-DISPLAYNAME-BRAND

Evidence: From: "PayPal" <security@xn--pypal-4ve.com>

Why it matters: Brand impersonation relies on recipients trusting the display name.

Limitations: Legitimate partners and resellers sometimes mention a brand in their sender name; protected-brand list is not exhaustive.

6. Limitations

- Headers can show that evidence supports or weakens a hypothesis; they cannot prove who wrote a message or whether an account was compromised.

- Received headers below the trust boundary are supplied by the sending side and may be forged.

- Authentication results are interpreted as recorded by the receiving server; signatures are not re-verified by this tool.

- Organisational domains use a built-in suffix list, not the full Public Suffix List; look-alike detection is heuristic and limited to protected domains.

- Message bodies, links and attachments are not analysed.

- Auto-trusted authserv-id 'mx1.uni-example.edu' (topmost Authentication-Results, matching the receiving server's domain).

7. Recommended next steps

- Do not act on requests in the message (payments, credentials, attachments) until the sender is verified out-of-band.

- Look up WHOIS / registration date of the look-alike domain and report it to the impersonated organisation.

- Correlate the origin IP and Message-ID with mail-server and gateway logs.

- Preserve this report together with the original file; verify the SHA-256 hash before any further processing.

Appendix A - Header fields as parsed (unfolded, in original order)

Control characters are shown as U+FFFD and very long values are truncated. The unmodified original is preserved separately under the SHA-256 above.

```
Received: from mx1.uni-example.edu (mx1.uni-example.edu [10.20.0.11]) by mailbox01.uni-example.edu with LMTP id
    8F2A1C; Mon, 14 Sep 2026 09:30:12 +0000
Authentication-Results: mx1.uni-example.edu; spf=pass smtp.mailfrom=xn--pypal-4ve.com; dkim=pass header.d=xn--
    pypal-4ve.com; dmarc=pass header.from=xn--pypal-4ve.com
Received: from mx.xn--pypal-4ve.com (mx.xn--pypal-4ve.com [45.137.22.15]) by mx1.uni-example.edu with ESMTPS id
    35856196 for <student@uni-example.edu>; Mon, 14 Sep 2026 09:30:00 +0000
Return-Path: <security@xn--pypal-4ve.com>
From: "PayPal" <security@xn--pypal-4ve.com>
To: <student@uni-example.edu>
Subject: Verify your identity
Date: Mon, 14 Sep 2026 09:29:00 +0000
Message-ID: <707213889216.816252@xn--pypal-4ve.com>
MIME-Version: 1.0
Content-Type: text/plain; charset=utf-8
```